Next Generation Data Centers and Energy Management Systems

Lecture 10A: Modeling and Simulation of an Energy Management System

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Berkeley
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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- **Modeling and Simulation of an Energy Management System**
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- ISO 50001 Energy Management Standard
- Implement a Convolution Neural Network (CNN)-Long Short Term Memory (LSTM) Architecture for load forecasting in a building energy management system

Typical layout of a data center facility

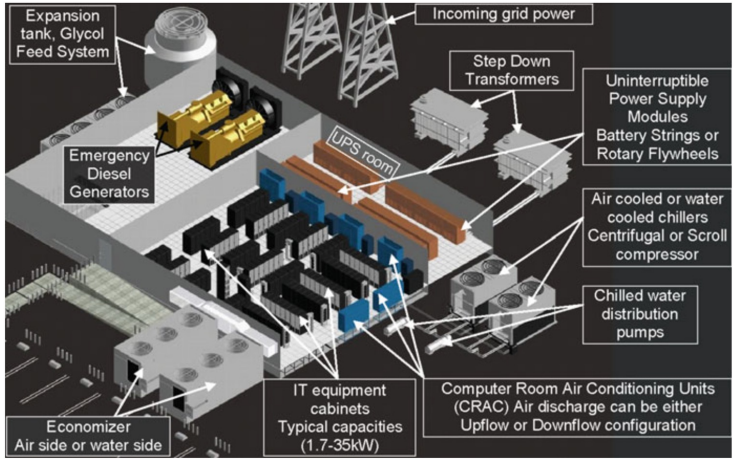


Figure 1: (Figure 1.1 in Ref. [2]).

Seminar 9

- Bryson Borzini
- Title: Building Next-Generation Data Centers: A Mechanical Contractor’s Perspective
- Date: November 17, 2026
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

ISO 50001 Energy Management Standard

- The International Organization for Standardization (ISO) has a standard, ISO 50001, which provides a framework for industrial facilities seeking to manage their energy use. Certification under this standard communicates how a company is tracking and reducing its energy use.
- Some Google-owned and operated data centers have this certification. Equinix claims to do this too.

CNN Deep Learning Model for Time Series Forecasting Problems [1]

- Neural networks that were designed to efficiently handle image data.
- They can automatically extract features from raw input data and can be applied to time series forecasting problems
- How to prepare time series data for CNNs and LSTMs?
- How to implement a CNN model for univariate time series forecasting problems?
- How to implement a CNN model for multivariate time series forecasting problems?

LSTM [1]

- The Long Short-Term Memory (LSTM) network is a type of recurrent neural network.
- Recurrent neural networks provide support for input sequence data.
- Instead of mapping inputs and outputs alone, they can learn a mapping function for the inputs over time to an output.
- They’ve been used for translation.
- They can automatically learn the temporal dependence of data.

CNN-LSTM [1]

Although CNNs can learn arbitrary mapping functions, the addition of support for input sequences in RNNs offers efficiency and greater performance for automatically learning temporal dependencies.
CNNs:

- They support efficient feature learning.
- Localized feature extractor; identifies short-term patterns (e.g., spike in IT load or drop in outside temperature).

LSTMs:

- They support efficient learning of temporal dependencies.
- Learns long-term time dependencies on those extracted features (e.g., daily or seasonal cooling trends).

CNN-LSTM

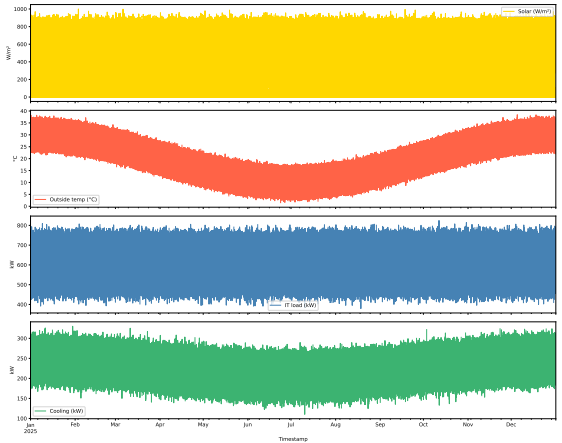


Figure 2: Raw time-series data.

References I

[1] Jason Brownlee.
Deep learning for time series forecasting: predict the future with MLPs, CNNs and LSTMs in Python.
Machine Learning Mastery, 2018.

[2] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.